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Task: Revision Tasks

Date: 10/05/2026

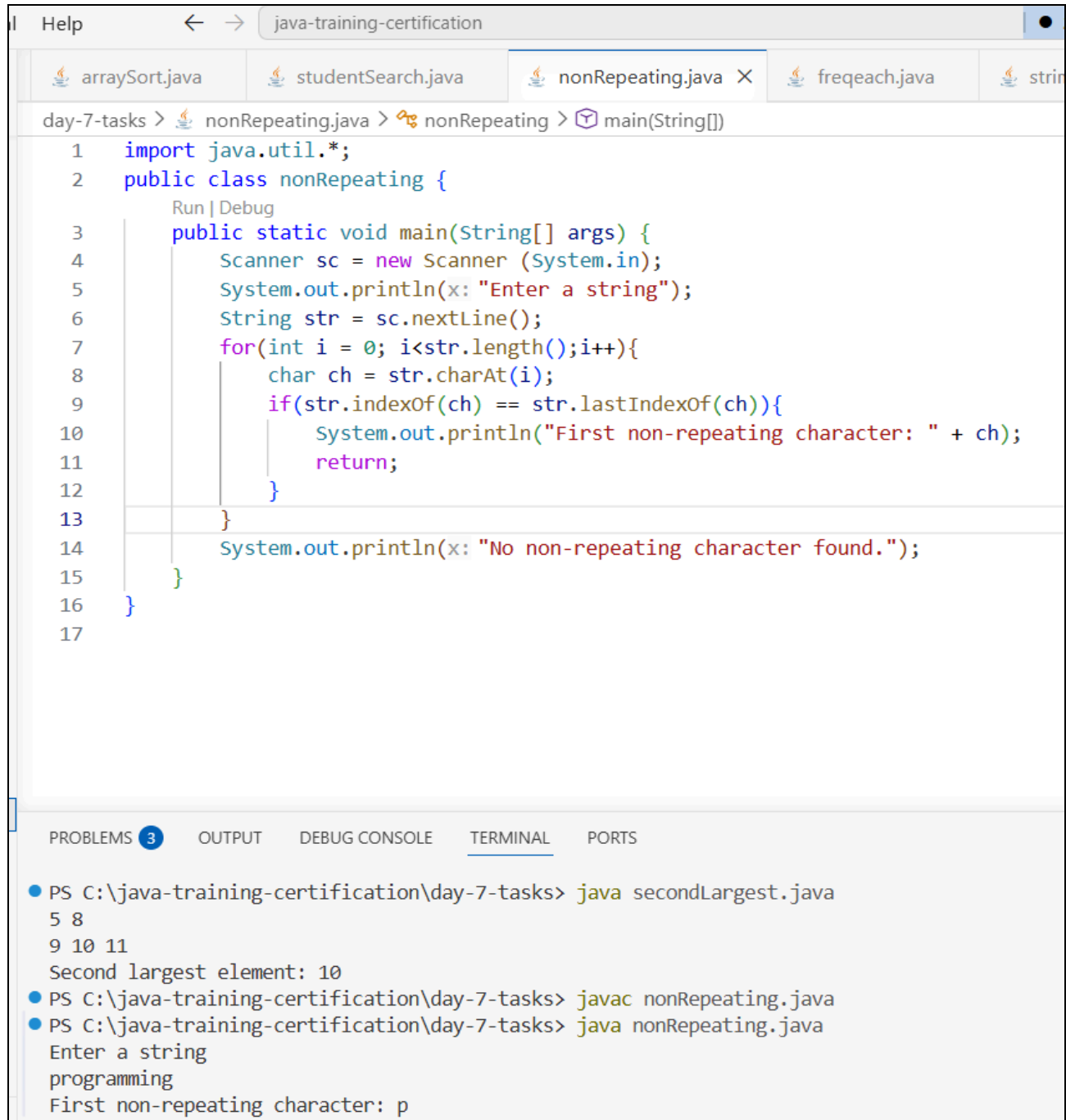
1. Second Largest Element in an Array

```
1  import java.util.*;
2  public class secondLargest {
3      public static void main(String[] args) {
4          Scanner sc = new Scanner(System.in);
5          int [] arr = new int[5];
6          System.out.println(x: "Enter 5 numbers: ");
7          for(int i = 0; i<arr.length;i++){
8              arr[i] = sc.nextInt();
9          }
10         int largest = arr[0];
11         int secondLargest = arr[0];
12         for(int i = 0; i<arr.length;i++){
13             if(arr[i]>largest){
14                 secondLargest = largest;
15                 largest = arr[i];
16             }
17             else if(arr[i]>secondLargest && arr[i]!=largest){
18                 secondLargest = arr[i];
19             }
20         }
21         System.out.println("Second largest element: " + secondLargest);
22     }
23 }
```

PROBLEMS 3 OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\java-training-certification\day-7-tasks> java bankacc.java
Current balance: 350.0
PS C:\java-training-certification\day-7-tasks> javac secondLargest.java
PS C:\java-training-certification\day-7-tasks> java secondLargest.java
5 8
9 10 11
Second largest element: 10
```

2. First Non-Repeating character in a String



The screenshot shows an IDE window titled "java-training-certification". The editor displays the file "nonRepeating.java" with the following code:

```
1 import java.util.*;
2 public class nonRepeating {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner (System.in);
5         System.out.println(x: "Enter a string");
6         String str = sc.nextLine();
7         for(int i = 0; i<str.length();i++){
8             char ch = str.charAt(i);
9             if(str.indexOf(ch) == str.lastIndexOf(ch)){
10                 System.out.println("First non-repeating character: " + ch);
11                 return;
12             }
13         }
14         System.out.println(x: "No non-repeating character found.");
15     }
16 }
17
```

The IDE interface includes tabs for "arraySort.java", "studentSearch.java", "nonRepeating.java", "fregeach.java", and "strin". The bottom panel shows the "TERMINAL" output:

```
PS C:\java-training-certification\day-7-tasks> java secondLargest.java
5 8
9 10 11
Second largest element: 10
PS C:\java-training-certification\day-7-tasks> javac nonRepeating.java
PS C:\java-training-certification\day-7-tasks> java nonRepeating.java
Enter a string
programming
First non-repeating character: p
```

3. Frequency Of Each Letter In A String

Help

← →

java-training-certification

arraySort.java

studentSearch.java

nonRepeating.java

freqeach.java >

day-7-tasks > freqeach.java > freqeach > main(String[])

1 import java.util.*;

2 public class freqeach {

3 Run | Debug

4 public static void main(String[] args) {

5 Scanner sc = new Scanner (System.in);

6 System.out.println(x: "Enter a string");

7 String str = sc.nextLine();

8 for(int i=0; i<str.length();i++){

9 char ch = str.charAt(i);

10 int count = 0;

11 for(int j = 0; j<str.length();j++){

12 if(ch == str.charAt(j)){

13 count++;

14 }

15 System.out.println(ch + " -> " + count + " times");

16 }

17 }

18 }

19

PROBLEMS 3

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

PS C:\java-training-certification\day-7-tasks> java stringComp.java

● PS C:\java-training-certification\day-7-tasks> javac freqeach.java

● PS C:\java-training-certification\day-7-tasks> java freqeach.java

Enter a string

112233

1 -> 2 times

1 -> 2 times

2 -> 2 times

2 -> 2 times

3 -> 2 times

3 -> 2 times

PS C:\java-training-certification\day-7-tasks>

4. Matrix multiplication

```
7-tasks / java-training-certification / day-7-tasks / matMul / matMul.java
import java.util.*;
public class matMul {
    Run | Debug
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.println(x: "Enter the number of rows for the first matrix:");
        int rows1 = sc.nextInt();
        System.out.println(x: "Enter the number of columns for the first matrix:");
        int cols1 = sc.nextInt();
        System.out.println(x: "Enter the number of rows for the second matrix:");
        int rows2 = sc.nextInt();
        System.out.println(x: "Enter the number of columns for the second matrix:");
        int cols2 = sc.nextInt();
        if (cols1 != rows2) {
            System.out.println(x: "Number of columns of the first matrix must be equal to the number of rows of the second matrix.");
            return;
        } else {
            int[][] matrix1 = new int[rows1][cols1];
            int[][] matrix2 = new int[rows2][cols2];
            int[][] result = new int[rows1][cols2];
            System.out.println(x: "Enter the elements of the first matrix:");
            for (int i = 0; i < rows1; i++) {
                for (int j = 0; j < cols1; j++) {
                    matrix1[i][j] = sc.nextInt();
                }
            }
            System.out.println(x: "Enter the elements of the second matrix:");
            for (int i = 0; i < rows2; i++) {
                for (int j = 0; j < cols2; j++) {
                    matrix2[i][j] = sc.nextInt();
                }
            }
            for (int i = 0; i < rows1; i++) {
                for (int j = 0; j < cols2; j++) {
                    result[i][j] = 0;
                    for (int k = 0; k < cols1; k++) {
                        result[i][j] += matrix1[i][k] * matrix2[k][j];
                    }
                }
            }
            System.out.println(x: "Result of matrix multiplication:");
            for (int i = 0; i < rows1; i++) {
                for (int j = 0; j < cols2; j++) {
                    System.out.print(result[i][j] + " ");
                }
                System.out.println();
            }
        }
    }
}
```

PROBLEMS 3 OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS C:\java-training-certification\day-7-tasks> java matMul.java

● PS C:\java-training-certification\day-7-tasks> java matMul.java

Enter the number of rows for the first matrix:

2

Enter the number of columns for the first matrix:

2

Enter the number of rows for the second matrix:

2

Enter the number of columns for the second matrix:

2

Enter the elements of the first matrix:

1 2

3 4

Enter the elements of the second matrix:

5 6

7 8

Result of matrix multiplication:

19 22

43 50

❖ PS C:\java-training-certification\day-7-tasks> █

5. String Compression

```
27 import java.util.*;
28 public class stringComp {
    Run | Debug
29     public static void main(String[] args) {
30         Scanner sc = new Scanner (System.in);
31         System.out.println(x: "Enter a string");
32         String str = sc.nextLine();
33         for(int i=0; i<str.length(); i++){
34             int count = 1;
35             while(i + 1 < str.length() && str.charAt(i) == str.charAt(i+1)){
36                 count++;
37                 i++;
38             }
39             System.out.print(str.charAt(i) + "" + count);
40         }
41     }
42 }
```

PROBLEMS 3 OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\java-training-certification\day-7-tasks> java stringComp.java
Enter a string
aabbccc
a2b2c3
PS C:\java-training-certification\day-7-tasks> 
```

6. Bank Account Simulation

```
day-7-tasks > bankacc.java > bankacc > display(String, double, double, double)
1  import java.util.Scanner;
2  public class bankacc {
    Run | Debug
3      public static void main(String[] args) {
4          Scanner sc = new Scanner(System.in);
5          System.out.println(x: "Enter the account holder's name:");
6          String name = sc.nextLine();
7          System.out.println(x: "Enter the initial balance:");
8          double balance = sc.nextDouble();
9          System.out.println(x: "Deposit amount:");
10         double deposit = sc.nextDouble();
11         System.out.println(x: "Withdrawal amount:");
12         double withdrawal = sc.nextDouble();
13         bankacc obj = new bankacc();
14         balance = obj.deposit(deposit, balance);
15         balance = obj.withdraw(withdrawal, balance);
16         obj.display(name, balance, withdrawal, deposit);
17     }
18     public double deposit(double amount, double balance){
19         balance += amount;
20         return balance;
21     }
22     public double withdraw(double amount, double balance){
23         if(amount > balance){
24             System.out.println(x: "Insufficient funds.");
25             System.out.println("Current balance: " + balance);
26             System.exit(status: 0);
27             return balance;
28         }
29         else{
30             balance -= amount;
31             return balance;
32         }
33     }
34     public void display(String name, double balance, double withdrawal, double deposit){
35         System.out.println("Account holder: " + name);
36         System.out.println("Current balance: " + balance);
37         System.out.println("Total deposit: " + deposit);
38         System.out.println("Total withdrawal: " + withdrawal);
39     }
}

PROBLEMS 3 OUTPUT DEBUG CONSOLE TERMINAL PORTS
● PS C:\java-training-certification\day-7-tasks> java bankacc.java
Enter the account holder's name:
ravi
Enter the initial balance:
500
Deposit amount:
200
Withdrawal amount:
800
Insufficient funds.
Current balance: 700.0
❖ PS C:\java-training-certification\day-7-tasks> 
```

7. Student Management using ArrayList

```
day-7-tasks > studentSearch.java > studentSearch
1  import java.util.*;
2  public class studentSearch {
    Run | Debug
3      public static void main(String[] args) {
4          Scanner sc = new Scanner(System.in);
5          ArrayList<String> sn = new ArrayList<>();
6          ArrayList<String> sr = new ArrayList<>();
7          ArrayList<String> sm= new ArrayList<>();
8          ArrayList<String> sd = new ArrayList<>();
9          System.out.println(x: "Enter the number of students:");
10         int n = sc.nextInt();
11         System.out.println(x: "Enter roll no to search:");
12         String searchRollNo = sc.next();
13         for(int i = 0; i < n; i++){
14             System.out.println("Enter the name of student " + (i+1) + ":");
15             sn.add(sc.next());
16             System.out.println("Enter the roll number of student " + (i+1) + ":");
17             sr.add(sc.next());
18             System.out.println("Enter the marks of student " + (i+1) + ":");
19             sm.add(sc.next());
20         }
21         sd.addAll(sr);
22         sd.addAll(sn);
23         sd.addAll(sm);
24         for(int i = 0 ; i < n; i++){
25             if(sd.get(i).equals(searchRollNo)){
26                 System.out.println(x: "Student found!");
27                 System.out.println("Name: " + sn.get(i));
28                 System.out.println("Roll Number: " + sr.get(i));
29                 System.out.println("Marks: " + sm.get(i));
30                 break;
31             }
32         }
33     }
34 }
```

PROBLEMS 3 OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\java-training-certification\day-7-tasks> java studentSearch.java
Enter the number of students:
2
Enter roll no to search:
1
Enter the name of student 1:
anu
Enter the roll number of student 1:
1
Enter the marks of student 1:
92
Enter the name of student 2:
bala
Enter the roll number of student 2:
2
Enter the marks of student 2:
85
Student found!
Name: anu
Roll Number: 1
```

8. Armstrong Numbers in Range

day-7-tasks >  armstrong_range.java >  armstrong_range

```
1  import java.util.*;
2  public class armstrong_range {
    Run | Debug
3  public static void main(String[] args) {
4      Scanner sc = new Scanner(System.in);
5      System.out.println(x: "Enter the starting number: ");
6      int start = sc.nextInt();
7      System.out.println(x: "Enter the ending number: ");
8      int end = sc.nextInt();
9      System.out.print("Armstrong numbers between " + start + " and " + end + " are:");
10     for(int i = start; i <= end; i++){
11         if(isArmstrong(i)){
12             System.out.print (i + " ");
13         }
14     }
15     System.out.println();
16 }
17 public static boolean isArmstrong(int num){
18     int sum = 0;
19     int temp = num;
20     while(temp != 0){
21         int digit = temp % 10;
22         sum += Math.pow(digit, 3);
23         temp /= 10;
24     }
25     return sum == num;
26 }
27 }
```

PROBLEMS  OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS C:\java-training-certification\day-7-tasks> java amstrong_range.java

● PS C:\java-training-certification\day-7-tasks> java armstrong_range.java

Enter the starting number:

100

Enter the ending number:

500

Armstrong numbers between 100 and 500 are:153 370 371 407

❖ PS C:\java-training-certification\day-7-tasks> 

9. Array Sorting without build-in methods

```
day-7-tasks > arraySort.java > arraySort > main(String[])
1  import java.util.*;
2  public class arraySort {
3      Run | Debug
4      public static void main(String[] args) {
5          Scanner sc = new Scanner (System.in);
6          int [] arr = new int[8];
7          for(int i = 0; i<arr.length;i++){
8              arr[i] = sc.nextInt();
9          }
10         int temp;
11         for(int i = 0; i<arr.length-1;i++){
12             for(int j = 0; j<arr.length-1-i;j++){
13                 if(arr[j]>arr[j+1]){
14                     temp = arr[j];
15                     arr[j] = arr[j+1];
16                     arr[j+1] = temp;
17                 }
18             }
19         }
20         System.out.print(s: "Sorted array: ");
21         for(int i = 0; i<arr.length;i++){
22             System.out.print(arr[i] + " ");
23         }
24         System.out.println();
25     }
26 }
```

PROBLEMS 3 OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\java-training-certification\day-7-tasks> java armstrong_range.java
Enter the ending number:
500
Armstrong numbers between 100 and 500 are:153 370 371 407
● PS C:\java-training-certification\day-7-tasks> java arraySort.java
5 9 10 11 51
15
88
33
Sorted array: 5 9 10 11 15 33 51 88
❖ PS C:\java-training-certification\day-7-tasks>
```

10. Division using Exception Handling

```
1  import java.util.*;
2  public class div {
3      Run | Debug
4      public static void main(String[] args) {
5          Scanner sc = new Scanner(System.in);
6          System.out.println(x: "Enter the first number:");
7          int num1 = sc.nextInt();
8          System.out.println(x: "Enter the second number:");
9          int num2 = sc.nextInt();
10         try{
11             int result = num1 / num2;
12             System.out.println("The result of division is: " + result);
13         } catch (Exception e) {
14             System.out.println("Error:"+ e.getMessage());
15         }
16         finally{
17             System.out.println(x: "Division operation completed.");
18         }
19     }
20 }
```

PROBLEMS 3 OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS C:\java-training-certification\day-7-tasks> java armstrong_range.java

Enter the ending number:

500

Armstrong numbers between 100 and 500 are:153 370 371 407

• PS C:\java-training-certification\day-7-tasks> java arraySort.java

5 9 10 11 51

15

88

33

Sorted array: 5 9 10 11 15 33 51 88

• PS C:\java-training-certification\day-7-tasks> javac div.java

• PS C:\java-training-certification\day-7-tasks> java div.java

Enter the first number:

15

Enter the second number:

0

Error:/ by zero

Division operation completed.

• PS C:\java-training-certification\day-7-tasks> █

